

# Hovercraft Hockey Hackathon

Build a puck that moves by barely touching the floor.

**Win condition:** Best glide plus control in a gym-safe tournament.

## THE SCIENCE YOU ARE USING

**Air cushion cuts friction.** A balloon pushes air down through a hole in a disc, lifting the disc on a thin cushion of air. With almost no contact, friction nearly vanishes, so a small push sends it gliding. Newton's laws then keep it moving until something stops it.

**Glide versus control.** Maximum glide is easy; control is hard. You balance lift and steerability to compete in gym-safe target hockey. Build quality and a clear explanation also score.

## HOW TO BUILD AND RUN IT

- 01 Mount the valve.** Ask a staff member to use low-temperature hot glue to attach a pop-top cap over the center hole of the disc and seal the edge.
- 02 Attach the balloon.** Stretch an inflated balloon over the closed cap so air escapes only through the hole.
- 03 Test the glide.** Open the cap just a little, set it down, and give a gentle push. A small opening glides longer and steers better; open wider only if it will not lift.
- 04 Practice control.** Learn to start and stop on target. Too much lift means no control.
- 05 Play target hockey.** Use the Glide Test and Target Rules sheet. Glide points are normalized to the field-best distance out of 30; divide the five-shot target total by 6 and round for a maximum of 25.

**Safety:** Low-temperature hot glue is staff-only; wait until it cools before handling the puck. Latex allergy note. No face-level launches. Play floor-only. A grown-up inflates the balloons; throw away any balloon that pops right away. Allergy: Latex balloon alternative: small bags or school-approved nonlatex balloons.

## MATERIALS

|                          |               |
|--------------------------|---------------|
| Recycled CDs or discs    | per team      |
| Pop-top bottle caps      | per team      |
| Balloons                 | per team      |
| Low-temperature hot glue | staff station |
| Targets                  | shared        |
| Floor tape               | shared        |

## YOUR PLAN AND DATA

Clean, complete data is worth as much as a fast result.

**Prediction (what will work best, and why):** \_\_\_\_\_

**The ONE variable we are testing:** \_\_\_\_\_

**Baseline result:** \_\_\_\_\_

**After redesign:** \_\_\_\_\_

## DEFEND YOUR DESIGN

**What evidence made you change your design?** \_\_\_\_\_

**If you ran it again, what is the ONE thing you would change?** \_\_\_\_\_

## HOW YOU SCORE (100 POINTS)

| SCORING CRITERION        | POINTS     |
|--------------------------|------------|
| Glide performance        | 30         |
| Target competition score | 25         |
| Explanation              | 20         |
| Redesign                 | 15         |
| Build quality            | 10         |
| <b>Total</b>             | <b>100</b> |